



CS102-Computer Programming 1

Programming Exercise 6

1st semester 2025-2026

Objective:

- To develop students' problem-solving and algorithmic thinking skills in programming.
- To apply user-defined functions and other fundamental programming concepts.

Prime and Perfect Numbers

Write a program that will continuously ask the user to input a positive integer number until the user inputs **EOF**. For each positive number, the program will determine and output on the screen whether it is a prime and/or a perfect number.

- a **prime number** is a whole number greater than 1 that has only two factors, i.e. 1 and the number itself. Prime numbers are divisible only by the number 1 or itself.
- a **perfect number** is a positive integer that is equal to the sum of its proper positive divisors, i.e. the sum of its positive divisors excluding the number itself.
 - Example: 6 is a perfect number. Its proper divisors are 1, 2, and 3, and $1 + 2 + 3 = 6$

The task of checking whether a number is a prime or perfect number should be implemented as two separate **functions** named ***isPrime()*** and ***isPerfect()*** with the positive **number** as their parameter. All the input and printing tasks should be handled by the ***main()*** function only, i.e. ***scanf*** and ***printf*** statements cannot be used in the function definition of the functions ***isPrime()*** and ***isPerfect()***. DO NOT use recursion in these functions.

Sample run output on the screen:

```
Enter a positive integer(Ctrl-Z to end): 10
prime: NO
perfect: NO

Enter a positive integer(Ctrl-Z to end): 13
prime: YES
perfect: NO

Enter a positive integer(Ctrl-Z to end): 6
prime: NO
perfect: YES

Enter a positive integer(Ctrl-Z to end): 17
prime: YES
perfect: NO

Enter a positive integer(Ctrl-Z to end): 28
prime: NO
perfect: YES

Enter a positive integer(Ctrl-Z to end): Ctrl-Z
End of program.Thanks!
```

Note to Students:

- Make sure to apply the concepts discussed in the lectures, especially on the user-defined functions.
- Be prepared to share and explain to the class your output on this exercise.
- Submit the source code with the filename: **LASTNAME-FIRSTNAME_PE6.c**